

Reverse Osmosis

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Abstract

Reverse osmosis has become very popular as a form of desalination. The process involves the use of a semi-permeable membrane with a pressure difference between both ends that pushes salt water through resulting in a freshwater permeate stream and a highly concentrated retentate stream. For this experiment different concentrated feed streams were tested at varying pressure differences. Using the experimental data it was found that the maximum recovery fraction was found to be 26% salinity. The rejection coefficient was found for each trial and the most efficient rejection coefficient was 96.6% for .11% salinity feed. The water permeability coefficient (A), and the salt permeability coefficient (B) were found to be $.02 \pm .01 m * s * bar^{-1}$ and $(4 \pm 2) * 10^{-3} kg * (m * s * bar)^{-1}$ respectively for the membrane used. Since only brackish water levels of salinity were used the results may not be valid for higher salinity levels. Also for the experiment only one membrane was used whereas in industry multiple membranes in series are used for higher levels of salt rejection.

1 Introduction

As the population grows there has been a scarcity of freshwater supplies due to an increased demand for consumption, use in industry and agriculture.¹ Water is abundant on earth, but mainly as seawater in our oceans. Water can be classified by its salinity into three main categories, seawater, brackish water, and freshwater. These classifications have a parts per million (PPM) concentration of 35,000 mg/L, 1,000-15,000, and 500 respectively. Only water with 500 mg/L or less is considered allowable for human consumption.² To combat the scarcity, reverse osmosis has been studied to find a practical way of desalination. Osmosis refers to the movement of water from an area of low to high concentration across a partially permeable membrane. Certain polymers are used for this process due to their structure which allows for water to pass through while dissolved substances do not.¹ The rejection is due to size and charge exclusion as well as the “physical-chemical interactions between solute solvent and membrane”.³ Reverse osmosis is pressure driven which allows the freshwater to be pushed out of the semi-permeable membrane, this results in two exit streams, one is a concentrated brine, the retentate and the other is almost pure water, the permeate.¹ Reverse osmosis has become very popular for desalination because it consumes less energy than other processes, which is about half compared to that needed for thermal processes. Other benefits are their simplicity, compact size, low investment costs and lower environment impact which allows for this process to be a sustainable option for water treatment.² Since the rise in popularity of reverse osmosis systems it has been found that performance is dependent on the feed conditions and the pressure across the membrane.

2 Background

Many different membrane-dependent desalination methods are commonly used on an industrial scale with reverse osmosis being one of the most favorable. Reverse osmosis typically operates on particle sizes that range between $5 \times 10^{-3} \mu\text{m}$ to $10^{-4} \mu\text{m}$, which is smaller than

other filtration methods. This allows reverse osmosis to be an effective in the removal of common chemical contaminants found in drinking water, such as aqueous salts and metal salts. It has also been found to be effective in the removal of certain strains of bacteria, viruses, and parasites. Its economic and relatively straight-forward design also makes it accessible for personal use.⁴ But there are a few notable disadvantages. Though reverse osmosis stands to be one of the most cost-effective water purification methods, there are still certain contents that cannot be removed, such as algae, microorganisms, or large suspended inorganic particles, which would require pre-treatment of the feed water stream. The presence of these particles poses potential complications such as membrane module and surface area blockage as well as membrane damage.⁵

The design for a spiral-wound reverse osmosis filtration process involves membrane sheets for filtration, feed channel spacer to create space between the membrane for feed fluid flow, and an additional spacer for filtered water, or permeate, flow. The three sheets are layered on top of one another and wrapped around a tube. The water feed is pumped through the feed channel spacer, which forces water through the membrane into the permeate channel. The permeate channel flows inward towards the tube and allows for the resulting permeate to be collected on the other end of the tube. The salty waste water, or retentate, is collected along the length of the membrane-wrapped tube and is usually recycled.⁶ Another design for membranes in reverse osmosis the method of hollow fine fiber, which is less common due to its ratio of permeate flow to membrane area being low compared the spiral-wound.⁵

3 Theory

Reverse osmosis systems rely heavily on the laws of mass transfer. Water flux,

$$J_w = A(\Delta p - \Delta \pi) = \frac{Q_p}{a} \quad (1)$$

where $\mathbf{A}$ is the water permeability coefficient, $\Delta\mathbf{p}$ is the operating pressure, and $\Delta\pi$ is the osmotic pressure. This relationship can also be expressed as the ratio between $\mathbf{Q_p}$ and $\mathbf{a}$, is permeate flow rate per membrane area and depends on operating conditions and the specific membrane. The water permeability coefficient,

$$A = \frac{D_w^M N_w^M V_w}{RTL} \quad (2)$$

where $\mathbf{D_w^M}$ is the diffusion coefficient of water inside the membrane, $\mathbf{N_w^M}$ is the solubility of water inside the membrane, $\mathbf{V_w}$ is the molar volume of water, $\mathbf{R}$ is the gas constant, $\mathbf{T}$ is the operating temperature, and $\mathbf{L}$ is the thickness of the separation layer, describes how easily pure water can move across the membrane. The water permeability coefficient is constant for a membrane and salt water concentration. Osmotic pressure,

$$\Delta\pi = \beta RT(C_f - C_p) \quad (3)$$

where β is the number of dissociated ions from the dissolved salt (which is assumed to be 1), $\mathbf{R}$ is the gas constant, $\mathbf{T}$ is the operating temperature, $\mathbf{C_f}$ is the concentration of salt in the feed water, and $\mathbf{C_p}$ is the concentration of salt in the permeate stream, is the pressure created when water molecules move across a semi-permeable membrane from the solution of lower to higher concentration. In the case of reverse osmosis, this pressure force must be opposed in order to produce a pure-water permeate stream and a salt-saturated retentate stream. This can be done by adjusting the operating pressure, $\Delta\mathbf{p}$, which is the pressure difference between the system and the atmosphere.

Another key governing equation describes the rate at which salt passes through the membrane, or the salt flux. The salt flux,

$$J_s = Br_j C_f = J_w C_p \quad (4)$$

where B is the salt permeability coefficient, C_f is the concentration of salt in the feed, and r_j is the rejection coefficient, is also directly correlated to water flux. The rejection coefficient,

$$r_j = \frac{C_f - C_p}{C_f} \quad (5)$$

describes the efficiency of separation of the membrane.

An ideal reverse osmosis system has large a value for water flux, J_w , and rejection coefficient, r_j . This result would signify that water can easily pass through membrane into the permeate stream while the salt is rejected and remains in the permeate stream. The factors that have commonly been observed to have the largest impact on these two parameters are operating pressure and feed water concentration.

4 Methods

The experiment was conducted using salinated water that was fed through a reverse osmosis membrane to produce retentate and permeate streams. The feed source was a tank that both the permeate and retentate streams were recycled back into continuously. The pressure difference was produced by closing a valve that controlled the outlet flow from the membrane. Both the permeate and retentate streams were measured using conductivity probes. A calibration curve was made using known concentrations to convert conductivity to salinity. For different trials, the concentration of salt in the feed was varied along with the pressure difference through the membrane. Experiments were run where the concentration of salt was varied from 1000-3600 ppm or .11-.31% salinity and the pressure was varied from 2-5 bar for each level of salinity. The pressure changed the flow rates of both streams, which was measured by timing the flow using a stopwatch and measuring the volume of water collected in that time period. The membrane had 11 layers each having a thickness of 6.25 mm and a length of 44.5 cm. In total 20 experiments were run at salinity levels of around 1,000, 1800, 2800, 3200, 3600 ppm at a varying pressure difference of 2, 3, 4, and 5 bar for each salinity level. For the high-

est salinity trial data could not be collected for the pressure being 2 bar since the flow of the permeate was not flowing fast enough.

5 Results and Discussion

Salty water was varied in 5 different concentrations to evaluate the performance of this reverse osmosis membrane in different conditions. The salty water was reduced from 0.11–0.31% in the feed stream, to 1.7–6.7 ppm in the retentate or purified stream. A calibration curve was used to convert conductivity measured from our equipment to the salinity of the water. This yielded coefficients of 0.35 ± 0.01 and 0.36 ± 0.02 , 0.37 ± 0.02 and 0.38 ± 0.02 for the permeate and retentate streams in the respective ranges of 0–20000, 0–2000 $\frac{\mu S}{cm}$ in Figs. 10 and 11.

5.1 Separation Efficiency

The recovery of the process is measured,

$$Y = \frac{Q_p}{Q_f} \quad (6)$$

where Y is the recovery fraction of the feed flow that passes through the membrane, Q_p is the permeate flow rate, and Q_f is the feed flow rate. The recovery fraction was found to be 0.11 ± 0.09 with a maximum of 26% across 0.11–0.31% salty water. The value of recovery fraction is not produced with confidence because the flow rates of the permeate and retentate streams were evaluated by measuring water collection over some duration of time, which is not quantifiable, due to the error source coming from human error. The recovery fraction is found to be the maximum at a pressure drop , $\Delta P = 5$ bar as shown in Fig. 1

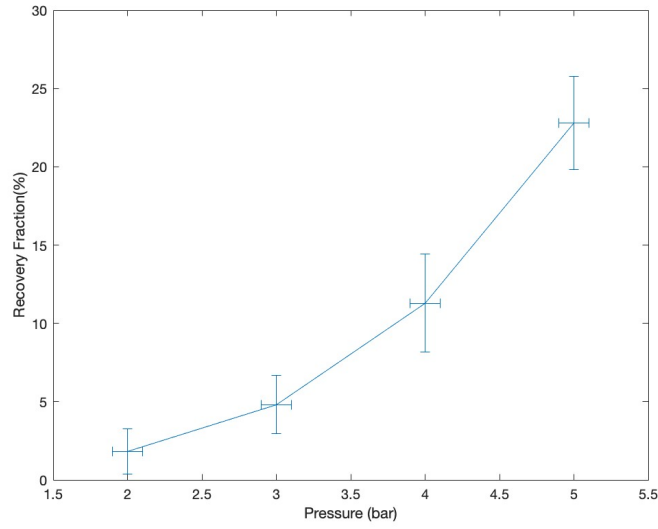


Figure 1: Recovery fraction averaged across salt water concentrations plotted along varying pressure drop

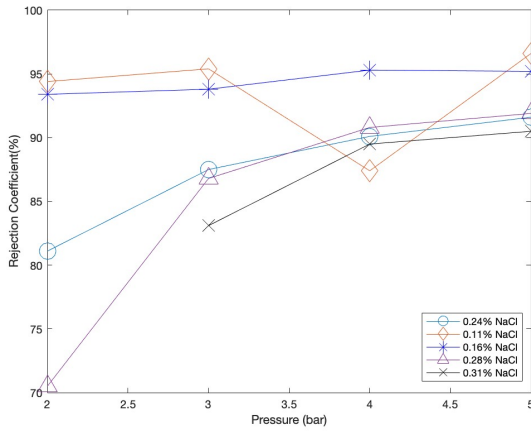


Figure 2: Rejection coefficient at different pressure driving force evaluated at different feed concentration in multiple trials

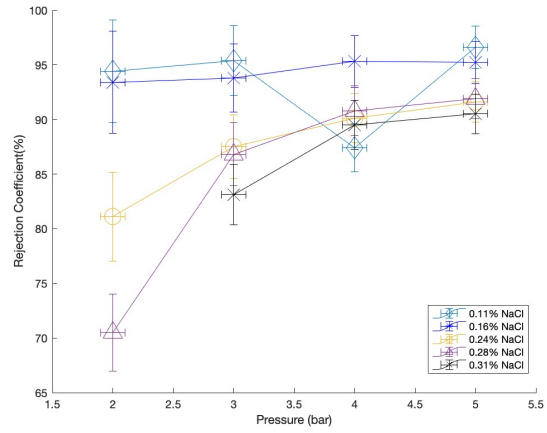


Figure 3: Rejection coefficient with propagated error

The rejection coefficient measured the separation efficiency of the RO membrane for the given solute. The highest rejection coefficient along various concentrations of NaCl, was found at a pressure drop, $\Delta P = 5$ bar as shown in Figs. 2 and 3. The 0.31% NaCl dataset in Fig. 3 only contains 3 data points since the value at 2 bar was omitted because the permeate flow rate was too low. This shows that higher concentrations give less recovery and separation, show-

ing that the membrane is limited and has reached its maximum salt flux, so it filters the salt water too slowly. This result shows the upper limitations in concentration to utilize for this experiment and its conditions. In [Fig. 3](#), it is more visible that a lower salt water concentration gives a higher rejection coefficient overall. This is because of the concentration driving force slightly increasing in the lower salt water concentrations. The larger effect is shown to be the pressure driving force, as the variation in rejection coefficient between salt water concentrations is much larger at lower pressure than higher pressure drops. [Fig. 2](#) shows a variation of 25% rejection at 2 bar, while it only varies by 6% rejection at 5 bar. This not only shows more precise separation at higher pressure, but more separation overall. Therefore, performance of this membrane is optimal at 5 bar. The most efficient rejection coefficients for this membrane ranged from 96.6% for 0.11% NaCl feed and 90.5% for 0.31% NaCl feed.

5.2 Permeability

The membrane is characterized by its water and salt permeability. The water and salt permeability can be evaluated with a linear fit across the varying pressure driving forces to return a value from the slope of each salty water concentration. This linear model relation is shown in [Fig. 4](#) and verified for a single concentration at different pressures. Using R-squared analysis in [Fig. 5](#), the linear relation is verified at different salt water concentrations, where the accuracy of the model increases towards 1 as concentration of NaCl is increased.

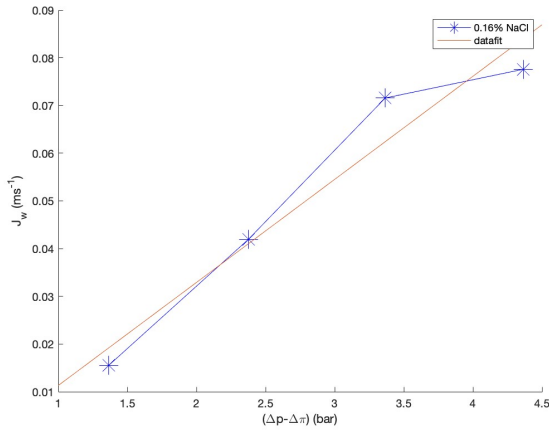


Figure 4: Linear fit of water permeability as the slope for water flux for a salt concentration of 0.16%

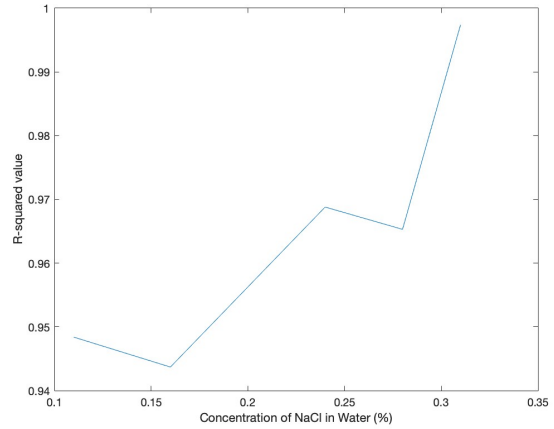


Figure 5: R-squared coefficients for varying salt concentrations

After verifying the linear model, the water permeability is measured along different concentrations in Fig. 6. The water permeability is reported for this RO thin film composite membrane as $0.02 \pm 0.01 \frac{m}{sbar}$. The error bars are propagated from the linear fit models of the calibration curve in Fig. 10 and linear regression model in Fig. 4. The fluctuations are due to the calibration curve which is unable to account for the measurement error in the conductivity probes since for every variation of value, probe setting, and probe location resulted in a different error. Changing between settings of the probes were mainly between the permeate and retentate stream, which used the 0–2000 and 0–20000 $\mu S/cm$ ranges respectively. The 0–200 $\mu S/cm$ range was omitted as the permeate stream would fluctuate above and below 200, despite giving us more accurate readings for conductivity under 200 $\mu S/cm$. In next attempts of this experiment, using probes using the same setting and precision will further help to decrease the error displayed in Fig. 6.

The salt permeability coefficient is also seen to be consistent over different concentrations of NaCl, reported to be a value of $(4 \pm 2) * 10^{-3} \frac{kg}{msbar}$

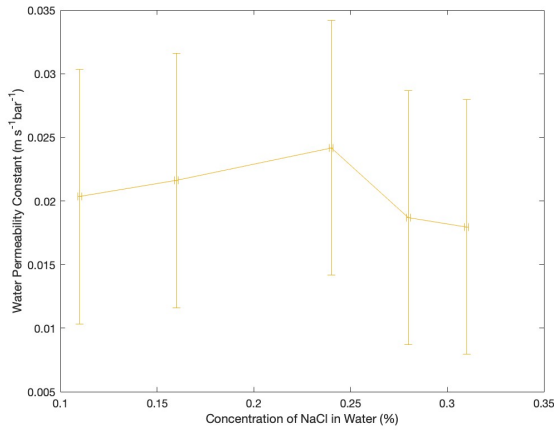


Figure 6: Water permeability coefficients over various NaCl concentrations

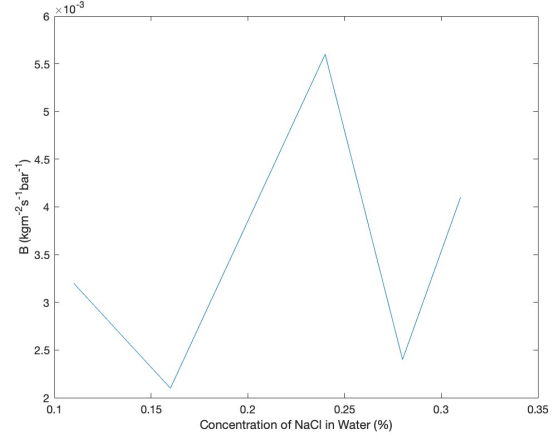


Figure 7: Salt permeability coefficients over various NaCl concentrations

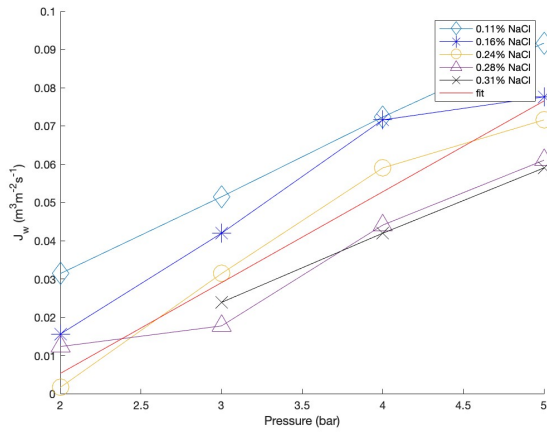


Figure 8: Water flux against pressure under various NaCl concentrations

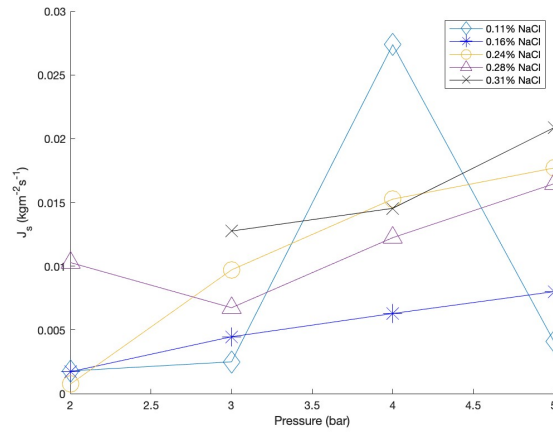


Figure 9: Salt flux against pressure under various NaCl concentrations

The water and salt flux are shown in Figs. 8 and 9, where the variation along pressure follows the slopes A and B, respectively. The outlier from the 0.11% concentration NaCl is seen in both figures Figs. 2 and 9 which can be due to non-uniform mixture when salinating the salt water tank, since it is the least concentrated trial done at the beginning.

6 Conclusions

In this experiment, values for the salt and water membrane permeability were found to be $.02 \pm .01 m * s * bar^{-1}$ and $(4 \pm 2) * 10^{-3} kg * (m * s * bar)^{-1}$ respectively. The maximum recovery fraction and rejection coefficient were found to be 26% and 96.9% respectively. Both of these coefficients were maximized at the lowest concentration of salt in the feed and the highest membrane pressure difference of .11% salinity and 5 bar. This shows that at lower the feed concentration and higher the pressure across the membrane is most efficient. Future experiments should investigate higher pressure differences and higher salinity feeds to see if the salt rejection coefficient, water and salt permeability coefficients are consistent at different conditions. These higher conditions should be further investigated anyways to see if reverse osmosis can be a viable option for desalinization since ocean water has a salinity much higher than what was tested. Also an experiment should also be done with two or more membranes in series to see if higher salt rejection can be achieved.

Bibliography

- (1) C. Fritzmann, e. a. Sate-of-the-art of reverse osmosis desalination. *Desalination* **2007**, 216, 1–76.
- (2) M.A. Alghoul, e. a. Review of brackish water reverse osmosis (BWRO) system designs. *Renew. Sustain. Energy Rev.* **2009**, 13, 2661–2667.
- (3) L. Malaeb, e. a. Reverse osmosis technology for water treatment: State of the art review. *Desalination* **2011**, 267, 1–8.
- (4) For Disease Control, C.; Prevention "A Guide to Drinking Water Treatment Technologies for Household Use".
- (5) El-Dessouky, H. T.; Ettouney, H. M., *Fundamentals of Salt Water Desalination*, 1st ed.; ELSEVIER: Amsterdam, The Netherlands, 2002, pp 410–452.

- (6) Shamraze Khan, e. a. Optimisation of design and operating parameters of reverse osmosis process for the removal of phenol from wastewater. *South African Journal of Chemical Engineering* **2023**, 43, 79–90.

Appendix

1. First main reference

- Author(s): Fritzmann, et al.
 - Year published: 2007
 - Journal name: Desalination
- (a) First Accomplishment: This article discusses reverse osmosis fundamentals such as membrane modules along with current and future developments
- (b) Second Accomplishment: This article explores some operational issues such as fouling and scaling.
- (c) Third Accomplishment: This article provides insights of necessary steps that are not included in the reverse osmosis system such as pre-treatment of the salty feed water and post-treatment of the concentrated brine that is expelled.

2. Second main reference

- Author(s): Alghoul, et al.
 - Year published: 2009
 - Journal name: Renew. Sustain. Energy Rev.
- (a) First Accomplishment: This article finds that adding an SWRO membrane at the second stage of the reverse osmosis process increases the recovery to about 83% and reduces boron concentration.
- (b) Second Accomplishment: This article finds that having a hybrid combination design of ultra low and conventional RO membranes will reduce energy consumption.

3. Third main reference

- Author(s): Malaeb, et al.

- Year published: 2011
 - Journal name: Desalination
- (a) First Accomplishment: This article reviews the recent advancements in reverse osmosis systems based on fouling, control techniques, and membrane characterization methods.
- (b) Second Accomplishment: This article explores issues with brine discharge and energy costs including energy recovery.

4. Fourth main reference

- Author: Center for Disease Control and Prevention
 - Year published: 2008
 - Journal name: Household Water Treatment
- (a) First Accomplishment: This fact sheet from the CDC identifies reverse osmosis to be the most effective method of drinking water purification in the various strains of bacteria, viruses, parasites as well as chemical contaminants when compared to micro-, ultra-, and nano-filtration.

5. Fifth main reference

- Author(s): Hisham T. El-Dessouky and Hisham M. Ettouney
 - Year published: 2002
 - Journal name: Fundamentals of Salt Water Desalination
- (a) First Accomplishment: This book explains the various methods of water desalination, with a focus on reverse osmosis. The pros and cons of reverse osmosis are explored in detail as techniques for minimizing the cons.

6. Sixth main reference

- Author(s): Shamraze Khan, Mudhar A. AL-OBAIDI, Chakib Kara-Zaitri, and Iqbal M. Mujtaba.
 - Year published: 2022
 - Journal name: South African Journal of Chemical Engineering
- (a) First Accomplishment: This article explores simulated optimization for spiral wound reverse osmosis using gPROMS software and found that the parameters that most governed performance were permeate concentration, solute flow, solute rejection, and water flux.
- (b) Second Accomplishment: This simulation results also showed that the removal of phenol from the waste-water stream was heavily influenced by the combination of membrane width, operating pressure, and feed temperature.

Calibration Curves

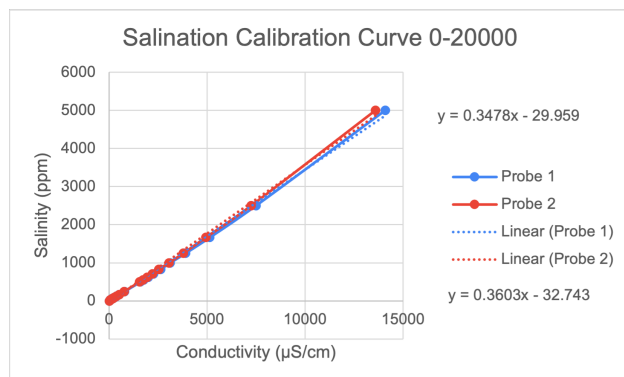


Figure 10: Calibration curve for range 0-20000 $\frac{\mu S}{cm}$ to convert from conductivity to salinity

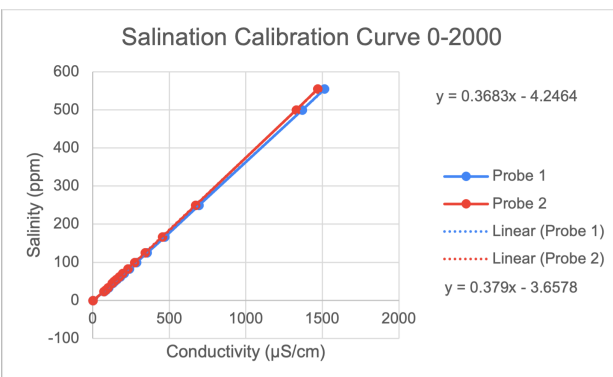


Figure 11: Calibration curve for range 0-2000 $\frac{\mu S}{cm}$ to convert from conductivity to salinity